

Building a Multi-node Hadoop Cluster on Ubuntu



This tutorial describes how to build a distributed Apache Hadoop multi-node cluster on four nodes running Ubuntu Server 14.04.1 LTS.

It is assumed that readers are aware of or have a good knowledge of Hadoop. This walkthrough is a guide to the configurations required to build a multi-node Hadoop cluster in which one node will serve as the NameNode and the others as DataNode.

The architecture of a Hadoop multi-node cluster is given in Figure 1.

Prerequisites to building the Hadoop cluster

Supported platforms: GNU and UNIX are supported as development and production platforms while Windows is only for development. For this tutorial, we will be using Ubuntu 14.04.1 (<http://www.ubuntu.com/download/server>).

Required software:

- Java is mandatory for running Hadoop. Do refer to the following guide prior to choosing the Java version: <http://wiki.apache.org/hadoop/HadoopJavaVersions>. For this tutorial, we will be using Java 7.

```
$apt-get install default-jre
```

```
$sudo apt-get install default-jdk  
$sudo apt-get update
```

- Download the latest Hadoop version from <http://www.apache.org/dyn/closer.cgi/hadoop/common/>
- ssh must be installed and sshd must be running to use Hadoop scripts:

```
$apt-get install ssh
```

User accounts and passwords for Hadoop installation

Adding a Hadoop user: Let us create a dedicated user account to run Hadoop. Though this is not mandatory, I would recommend it as it helps in isolating the Hadoop installation from other software applications and other user accounts running on the same node.

```
$sudo addgroup hadoop #this will add Hadoop group, make sure  
that it is done in all the nodes
```